

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD):

FR-DLT-001: Load Customer and Order Data into Delta Tables

- Title: Load Customer and Order Data into Delta Tables
- Description: The system must read source CSV data from a volume and load it into Delta tables for customer and order data.
- Preconditions:
 - CSV files are available at the specified locations (/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata and /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata)
- Delta tables for customer and order data do not exist or are empty
- Main Flow / Functional Steps:
 - Read customer CSV data from /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata into a Delta table named "customer"
 - Read order CSV data from /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata into a Delta table named "order"
 - Validate that the schema of the customer table matches the expected schema (CustId, Name, EmailId, Region)
 - Validate that the schema of the order table matches the expected schema (OrderId, ItemName, PricePerUnit, Qty, Date, CustId)

FR-DLT-002: Transform Order Data by Adding TotalAmount Column

- Title: Transform Order Data by Adding TotalAmount Column
- Description: The system must add a new column "TotalAmount" to the order table, which is the product of PricePerUnit and Qty.
- Preconditions:
 - Order Delta table exists and is not empty
- Main Flow / Functional Steps:
 - Add a new column "TotalAmount" to the order table
 - Calculate the value of TotalAmount as the product of PricePerUnit and Qty

FR-DLT-003: Cleanse Customer and Order Data

- Title: Cleanse Customer and Order Data
- Description: The system must remove null and duplicate records from both customer and order tables.
- Preconditions:
- Customer and order Delta tables exist and are not empty
- Main Flow / Functional Steps:
- Remove rows with null values from both customer and order tables
- Remove duplicate rows from both customer and order tables

FR-DLT-004: Create ordersummary Table and Load Joined Data

- Title: Create ordersummary Table and Load Joined Data
- Description: The system must create an ordersummary table if it does not exist and load joined data from customer and order tables into it.
- Preconditions:
- Customer and order Delta tables exist and are not empty
- ordersummary table does not exist or is empty
- Main Flow / Functional Steps:
- Create ordersummary table with the expected schema (CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date) if it does not exist
- Join customer and order tables on CustId
- Load the joined data into the ordersummary table as an SCD type 2 table

FR-DLT-005: Update ordersummary SCD Type 2 Table on Customer Changes

- Title: Update ordersummary SCD Type 2 Table on Customer Changes
- Description: The system must update the ordersummary SCD type 2 table whenever there is a change in the customer table.
- Preconditions:
- ordersummary table exists and is not empty
- Customer table exists and is not empty
- Main Flow / Functional Steps:
- Detect changes in the customer table
- Update the ordersummary table by making old records inactive and new records active

- Update StartDate and EndDate accordingly to maintain history

FR-DLT-006: Create customeraggregatespend Table and Load Aggregated Data

- Title: Create customeraggregatespend Table and Load Aggregated Data
- Description: The system must create a customeraggregatespend table if it does not exist and load aggregated data from the ordersummary table into it.
- Preconditions:
 - ordersummary table exists and is not empty
 - customeraggregatespend table does not exist or is empty
- Main Flow / Functional Steps:
 - Create customeraggregatespend table with the expected schema (Name, TotalAmount, Date) if it does not exist
 - Aggregate TotalAmount from ordersummary table by Name and Date
 - Load the aggregated data into the customeraggregatespend table

FR-DLT-007: Implement Data Processing using Delta Live Tables

- Title: Implement Data Processing using Delta Live Tables
- Description: The system must implement the data processing pipeline using Delta Live Tables.
- Preconditions:
 - All previous steps have been executed successfully
- Main Flow / Functional Steps:
 - Implement the data processing pipeline using Delta Live Tables to perform the required data transformations and loading.